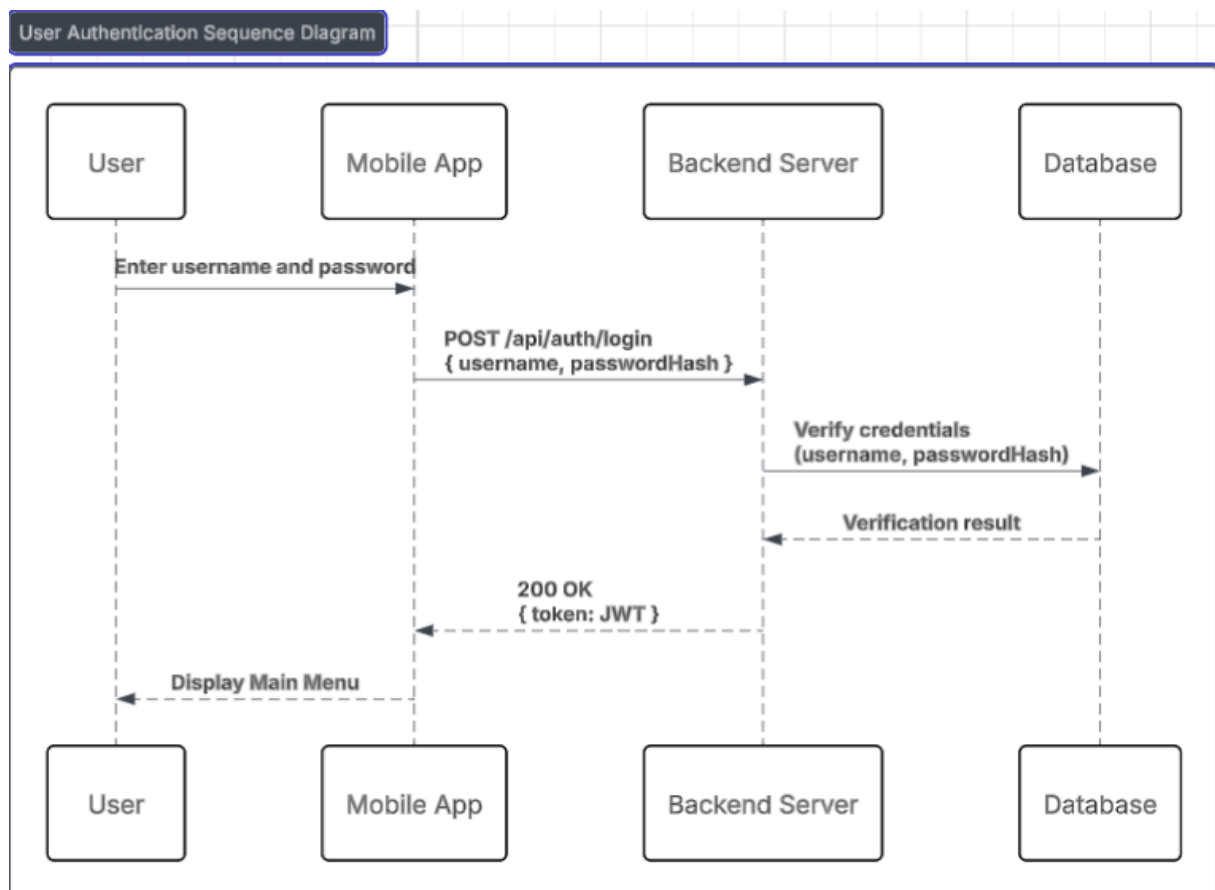


Sequence Diagrams

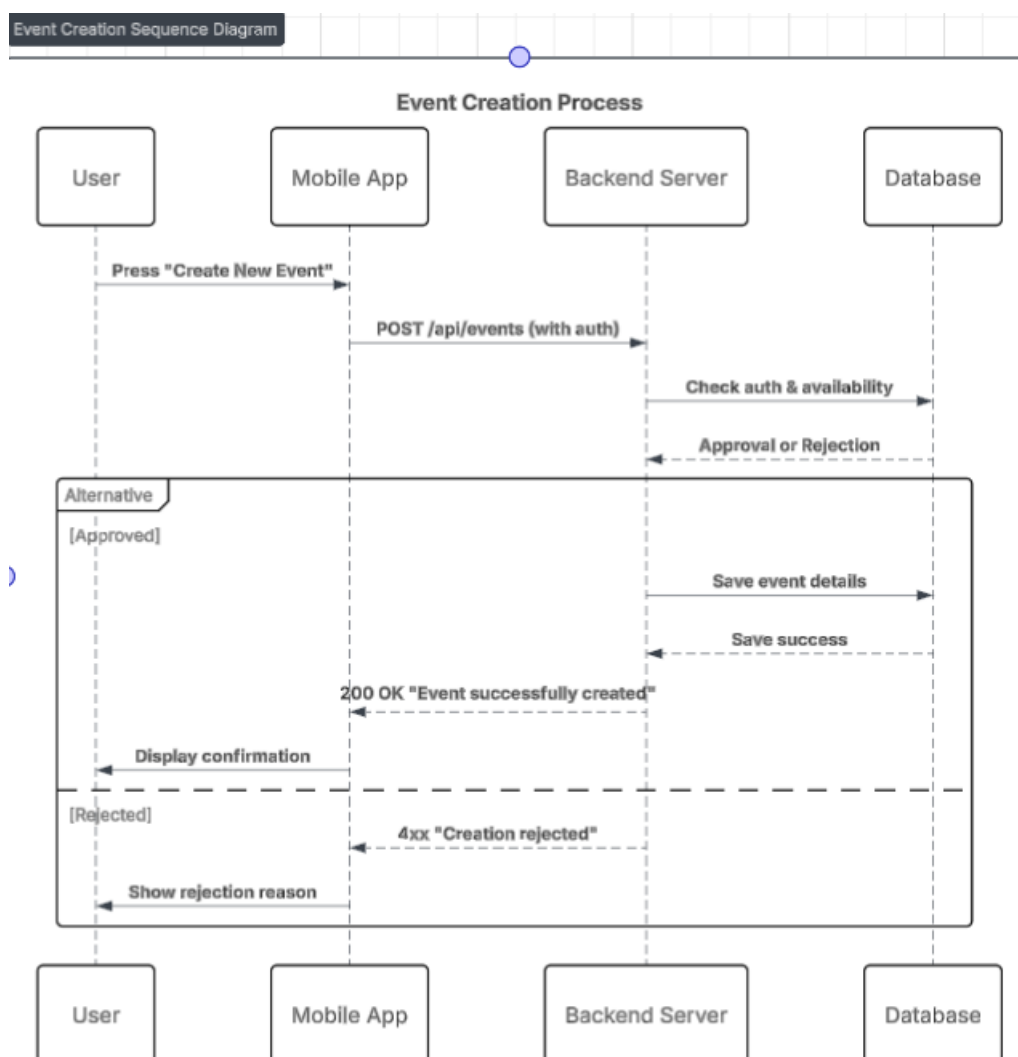
User Authentication Diagram

This sequence diagram illustrates the User Authentication process, focusing on the steps required to establish a secure session. The User initiates the flow by entering their credentials into the Mobile App. The app then sends a POST /api/auth/login request, typically containing a hashed password, to the Backend Server. The server verifies these credentials against the Database. If successful, the Backend Server sends a 200 OK response back to the Mobile App containing a JSON Web Token (JWT), which the app uses for secure subsequent requests. Finally, the Mobile App uses the granted access to Display the Main Menu to the User.



Event Creation Sequence Diagram

This sequence diagram provides a concise summary of the Event Creation Process within the application. The flow begins when the User initiates the creation from the Mobile App, which then sends an authenticated API request to the Backend Server. The server's immediate responsibility is to check both user authentication and availability against the system's records. The process follows an Alternative path: if the checks are successfully [Approved], the event details are saved in the Database, and the user receives a success confirmation (200 OK). Conversely, if the checks [Rejected] due to security or scheduling conflicts, the server returns an error (4xx), and the Mobile App displays the reason for rejection to the User.



Event Search Diagram

This sequence diagram outlines the Event Search and Recommendation process. The flow begins when the User selects search filters and the Mobile App sends a GET request to the Backend Server. The server delegates the heavy lifting to the Matching Engine, which queries the Database for user availability and active events. After receiving the data, the Matching Engine processes it to create a ranked and filtered event list, which is then returned to the Backend Server. Finally, the Backend Server sends a 200 OK response containing the list to the Mobile App for display to the User.

